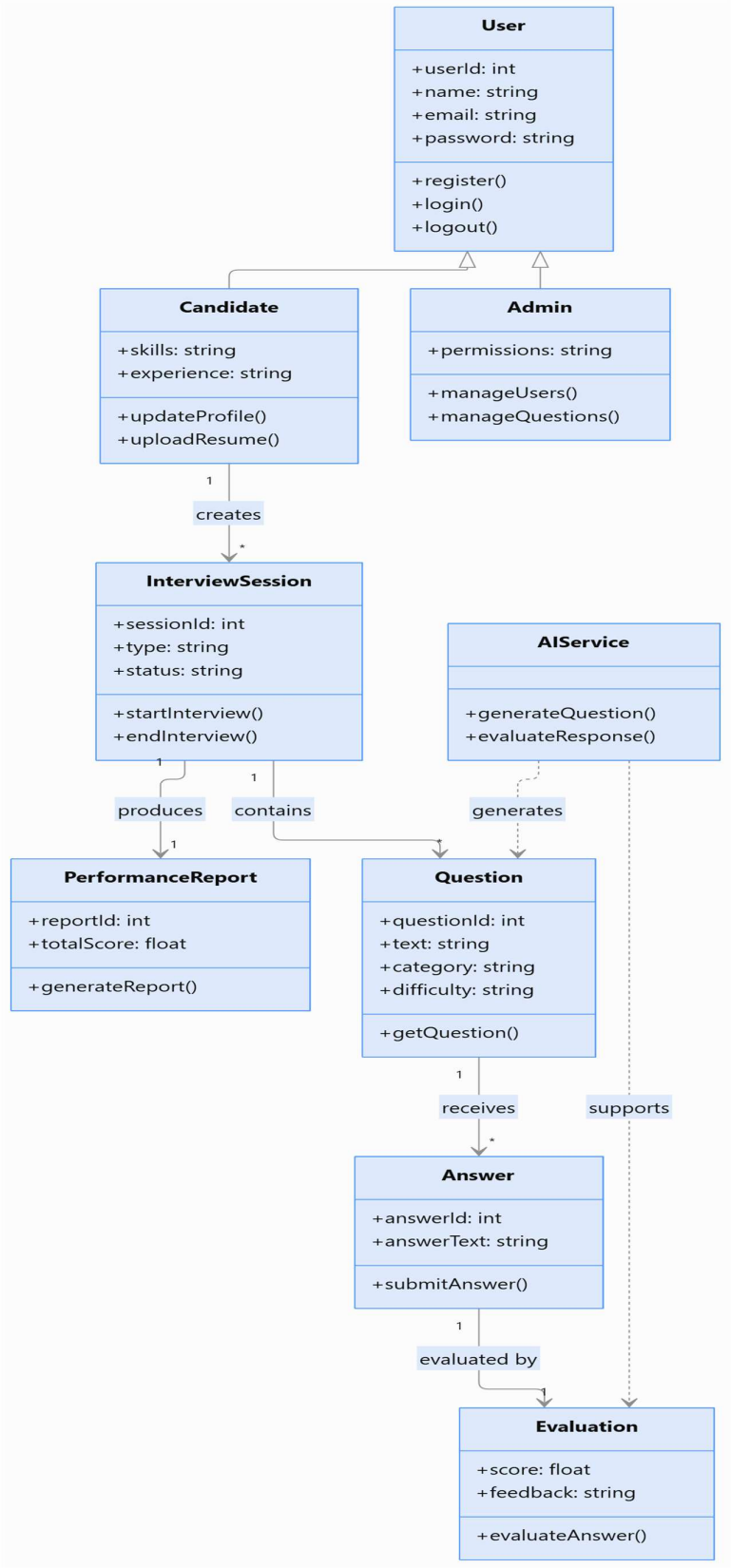
Lab 6: UML Class Diagram Design

Aim: To design a UML Class Diagram representing the structure of the AI-Based Interview Preparation Platform.

1. Classes and their responsibilities

Class	Attributes	Methods
User	userId, name, email, password	register(), login(), logout()
Candidate	resume, skills, experience	updateProfile(), uploadResume()
InterviewSession	sessionId, type, date, status	startInterview(), endInterview()
Question	questionId, text, category, difficulty	getQuestion()
Answer	answerId, answerText, timestamp	submitAnswer()
Evaluation	score, feedback, strengths, weaknesses	evaluateAnswer(), generateFeedback()
AIService	modelName, apiKey	generateQuestion(), evaluateResponse()
PerformanceReport	reportId, totalScore, date	generateReport(), viewReport()
Admin	adminId, permissions	manageUsers(), manageQuestions()

2. UML Class Diagram



3. Explanation

- User: Common base class containing login and registration details.
- Candidate: Student or job seeker who practices interviews.
- Admin: Manages users and the question bank.
- InterviewSession: Stores interview type, status and session information.
- Question and Answer: Represent interview questions and candidate responses.
- Evaluation: Stores scores and feedback generated from answers.
- AIService: Integrates an external AI API or a locally hosted model.
- PerformanceReport: Summarizes the candidate's interview performance.

Result: The UML Class Diagram represents the static structure and relationships of the proposed system.